

MITCHELL COHEN

Montreal, QC, Canada

(514) 661-3427 ♦ mitchell.cohen3@mail.mcgill.ca ♦ <https://mitchellcohen3.github.io/>

Robotics PhD candidate specializing in state estimation for GPS-denied systems with 4+ years of industry collaboration experience. Strong foundation in estimation theory and practical expertise developing real-time algorithms for autonomous systems. Co-authored 9 peer-reviewed articles in robotics conferences and journals with over 200 citations.

EDUCATION

Doctor of Philosophy in Mechanical Engineering *2020 - 2026 (expected)*

McGill University, Montreal, QC

Advancing Methods for Consistent and Efficient Visual-Inertial Navigation

Master of Mechanical Engineering *2018 - 2020*

McGill University, Montreal, QC

Specialization in dynamics, control, and estimation of aerial autonomous systems. GPA: 3.83/4.00

Bachelor of Mechanical Engineering *2014 - 2018*

McGill University, Montreal, QC

Graduated with distinction. GPA: 3.72/4.00

TECHNICAL STRENGTHS

State Estimation

- Derivation and implementation of optimization and filtering-based Bayesian state estimators (EKF, UKF, MAP estimators, sliding-window filters) for real-time, multi-sensor robotic navigation systems.
- Knowledge and implementation of common state estimation techniques for visual-inertial platforms, including IMU preintegration, the multi-state constraint Kalman filtering framework, methodologies for consistent estimator design, and online spatiotemporal calibration of multi-sensor platforms.
- Experience with libraries and frameworks commonly used in robotic state estimation, including Eigen, Ceres, GTSAM, and OpenCV, as well as popular visual-inertial algorithms such as VINS-Mono, OpenVINS, and ORB-SLAM3.

Control and Guidance of Aerospace Systems

- Knowledge and simulation experience of control and guidance strategies for both fixed-wing and rotorcraft UAV systems.

Dynamic Simulation of Multibody Systems

- Experience with derivation and simulation of equations of motion of 6 DoF robotic systems.

RELEVANT EXPERIENCE

DENSO Corporation *2020 - 2024*

Research Collaborator

- Researched, developed and implemented real-time visual-inertial odometry algorithms for automotive applications as part of a collaborative research project between DECAR at McGill University and DENSO Corporation.

- Conducted a 3-month internship (Sept. 2023 - Dec. 2023) at the DENSO R&D office in Tokyo, Japan, focused on simultaneous visual-inertial estimation and tracking of moving objects.

ARA Robotique

2019 - 2020

Research Intern

- Guidance, navigation, and control research intern for unconventional vertical takeoff and landing (VTOL) unmanned aerial vehicle (UAV).
- Derivation and analysis of linear control allocation techniques for unconventional forward-flight aircraft.

Summer Undergraduate Research In Engineering

Summer 2018

McGill University

- Dynamic modeling, simulation, and control of a passively-coupled tilt-rotor VTOL UAV.

SKILLS

Programming Languages	C++, Python, MATLAB
Libraries & Frameworks	Eigen, Ceres, GTSAM, OpenCV, ROS1/ROS2
Tools	Linux, Git, Docker
Languages	English, French

PUBLICATIONS

- **M. Cohen**, V. Korotkine, and J. R. Forbes, “Observability and Consistency Analysis for Visual-Inertial Navigation with Anchored Feature Parameterizations”, to appear at the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- V. Korotkine, **M. Cohen**, and J. R. Forbes, “Globally Optimal Data-Association-Free Landmark-Based Localization Using Semidefinite Relaxations”, in *Robotics and Automation Letters (RA-L)*, 2025.
- V. Korotkine, **M. Cohen**, and J. R. Forbes, “A Hessian for Gaussian Mixture Likelihoods in Nonlinear Least Squares”, in *Robotics and Automation Letters (RA-L)*, 2024.
- C. C. Cossette, **M. Cohen**, V. Korotkine, A. D. C. Bernal, M. A. Shalaby, and J. R. Forbes, “navlie: A Python Package for State Estimation on Lie Groups”, in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- D. Lisus, **M. Cohen**, and J. R. Forbes, “Know What You Don’t Know: Consistency in Sliding Window Filtering with Unobservable States Applied to Visual-Inertial SLAM”, *IEEE Robotics and Automation Letters (RA-L)*, 2023.
- **M. Cohen**, and J. R. Forbes “Navigation and Control of Unconventional VTOL UAVs in Forward-Flight with Explicit Wind Velocity Estimation”, *Robotics and Automation Letters (RA-L)*, 2020.
- N. van Der Laan, **M. Cohen**, J. Arsenault, and J. R. Forbes, “The Invariant Rauch-Tung-Striebel Smoother”, *Robotics and Automation Letters (RA-L)*, 2020.
- **M. Cohen**, K. Abdulrahim, and J. R. Forbes, “Finite-Horizon LQR Control of Quadrotors on $SE_2(3)$ ”, *Robotics and Automation Letters (RA-L)*, 2020.
- R. Chiappinelli, **M. Cohen**, M. Doff-Sotta, M. Nahon, J. R. Forbes, and J. Apkarian, “Modeling and Control of a Passively-Coupled Tilt-Rotor Vertical Takeoff and Landing Aircraft”, *International Conference on Robotics and Automation (ICRA)*, 2019.